

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	(i) States both hypotheses correctly	AO2.5	B1	$H_0: \mu = 3$ $H_1: \mu > 3$
	Finds both $\bar{x}$ and s^2 or s (condone n divisor if retrieved in formula)	AO1.1b	B1	$\bar{x} = \frac{28.8}{9} = 3.2$ $s^2 = \frac{\sum (x - \bar{x})^2}{n-1} = 0.075$ $(s = 0.27386)$
	Uses formula to find test statistic PI	AO1.1a	M1	$t = \frac{3.2 - 3}{\sqrt{\frac{0.075}{9}}}$ Test statistic
	States test statistic correctly (AWFW 2.15 – 2.23) or finds p	AO1.1b	A1	Test statistic $t = 2.19$ $p = 0.020(4)$
	Evaluates t model by comparing test statistic and correct critical value Or by comparing p -value with 0.05	AO3.5a	M1	Critical value for 1-tail test at the 5% level of significance for a t_8 distribution is 1.8595, $2.19 > 1.8595$. Or $0.020(4) < 0.05$
	Infers that H_0 should be rejected	AO2.2b	E1	Reject H_0
	Completes test correctly and gives conclusion in context. (conclusion should not be definite)	AO3.2a	E1	As $2.19 > 1.8595$ The evidence suggests that the impurity, per cent, level in the chemical is too high
(ii)	Identifies the limitation of the t -distribution model if data not normally distributed	AO3.5b	E1	Assumption: the level of impurity is normally distributed
(b)	Makes statement (could be implied by quoting the	AO3.5c	E1	It would be a z-test rather

critical z-value) that z-test would be required			than a t-test [critical value change to $z = 1.645$]
Mentions change to σ or change to test statistic	AO2.4	E1	Would use $\sigma = 0.25$ not $\sigma = 0.27386$ or Value of ts changes to 2.4
Total 10 marks			

Q2.

(a) $H_0: \mu = 175$

$H_1: \mu < 175$

Both; accept $H_0: \mu \geq 175$

*Do not accept mean or $\bar{x}$
but accept population mean*

B1

$\bar{x} = 168.1$

B1

$$z = \frac{168.1 - 175}{9.4 / \sqrt{6}}$$

For use of $9.4 / \sqrt{6}$

M1

For rest of formula (ignore sign)

m1

$= -1.798$

Must be negative AWRT -1.80

A1

$CV = -1.6449$

AWFW -1.64 to -1.65

B1

$-1.6449 > -1.798$ so test statistic in critical region

*Comparison of correct test statistic with correct CV
must be seen (diagram or words)*

Reject H_0 , significant evidence that batch **mean** is less than 175 grams

OE; suspicion supported
Must be in context AG

A1

7

(b) $H_0: \mu = 175$

$H_1: \mu < 175$

Award B1 for both correct if not scored in (a)

$$t = \frac{169.4 - 175}{11.2 / \sqrt{20}}$$

For use of 11.2 / $\sqrt{20}$

M1

For rest of formula (ignore sign)

m1

$= -2.236$

Must be negative AWRT - 2.24

A1

$CV(t_{19}) = -2.539$

AWRT - 2.54

B1

$-2.236 > -2.539$ so test statistic not in critical region

*Comparison of correct test statistic with correct CV
(need not be seen)*

Accept H_0 , no significant evidence that batch mean / weight is less than 175 grams

OE; suspicion not supported

A1

5

(c) Because the significance level is 1% instead of 5%

OE; eg SL is different

Reference to sample size $\Rightarrow$ E0

E1

1

[13]

Q3.

Assume that lengths of shots are normally distributed

$$s_n^2 = 124; s_n = 11.1$$

$$\text{iff } \frac{s_n}{3} \text{ used}$$

B1

$$\bar{x} = 184$$

$$s^2 = \frac{1240}{9} = 137.7 \quad (s = 11.7)$$

CAO

AWFW 137.7 to 138

both $\bar{x}$ and s^2 (or s)

B1

$$H_0 : \mu = 190$$

$$H_1 : \mu \neq 190$$

Both

B1

$$t = \frac{184 - 190}{\sqrt{1240/9 \times 10}}$$

$$t = \frac{\text{their } \bar{x} - 190}{\frac{\text{their } s_{n-1}}{\sqrt{10}}}$$

$$\text{or } \frac{\text{their } \bar{x} - 190}{\frac{\sqrt{9}}{\text{their } s_n}}$$

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M1

$$t = -1.62$$

AWRT

A1

$$v = 9 \Rightarrow t_{\text{crit}} = \pm 2.821$$

(accept 2.82)

B1

$$-2.821 < -1.62 < 2.821 \text{ accept } H_0$$

Evidence to support Lorraine's belief at 2% level of significance

E1

Q4.

$$\bar{x} = 82; s^2 = 31.1 (s = 5.58)$$

B1B1

Assumption:

The number of customers served daily at the post office counter forms a Normal distribution.

B1

$$H_0 : \mu = 79$$

$$H_1 : \mu > 79$$

B1

$$t = \frac{82 - 79}{5.58 / \sqrt{12}}$$

$$\frac{\text{their } \bar{x} - 79}{\text{their } s / \sqrt{12}}$$

M1

$$t = 1.86$$

(AWRT)

A1

$$v = 11 \Rightarrow t_{\text{crit}} = 1.796$$

B1

Reject H_0

A1

Sufficient evidence at 5% level of significance to support Judith's belief.

$$\text{Iff } t_{\text{calc}} > t_{\text{crit}}$$

E1

9

[9]

Q5.

(a) (i) $\bar{x} = 43.5$

B1

$$s = 2 (s^2 = 4)$$

B1

Assumption: Weights of boxes are normally distributed

B1

$$t_{0.975} = 2.365$$

B1

95% CI for μ :

$$\left. \begin{array}{l} 43.5 \pm 2.365 \times \frac{2}{\sqrt{8}} \\ 43.5 \pm 1.6723 \end{array} \right\}$$

M1

$$\Rightarrow (41.8, 45.2)$$

(AWRT)

A1

6

(ii) CI contains mean (45)

*Must be clear use of 45 and **not** 43.5*

B1dep

Bishen's belief probably justified

B1dep

or

[Since 45 within CI] but close to upper limit, there is some evidence that Bishen's Belief is untrue
[but the evidence is not significant at 5 %.]
(75% of sample less than 45 grams)

(B1)

2

- (b) (i) $H_0 : \mu = 45$
 $H_1 : \mu < 45$
(both)

B1

$$t = \frac{43.5 - 45}{\frac{2}{\sqrt{8}}}$$

Test statistic:

M1

$$= -2.12$$

$$P(t_7 < -2.12) = 0.035791 < 0.05$$

A1

$$v = 7 \Rightarrow t_{\text{crit}} = -1.895$$

B1

$\Rightarrow$ Reject H_0

A1

Evidence at the 5% level of significance to support
Abi's claim that **mean** content < 45 grams

E1

6

(ii) Type I error

B1

have/may have rejected H_0 when H_0 true
Clear statement

B1

or
No error

(B1)

have/may have accepted H_0 when H_0 true
Clear statement

(B1)

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2

[16]

Q6.

$$\bar{x} = 4.1 \quad s = 0.392 \quad (s^2 = 0.153)$$

both

B1

$$H_0 : \mu = 3.8$$

$$H_1 : \mu > 3.8$$

both

B1

$$t = \frac{4.1 - 3.8}{\frac{0.392}{\sqrt{7}}} = 2.03$$

M1A1

$$t_{\text{crit}} = 1.943$$

B1ft

Reject H_0

A1

Evidence at 5% level of significance to support the doctor's belief that the cholesterol level is higher than the management's claim of 3.8

E1

Cholesterol levels normally distributed

B1

8

[8]

Q7.

(a) $H_0 : \mu = 30$
 $H_1 : \mu > 30$

B1

$$\bar{x} = 33.5 \quad \text{and} \quad s = 4.25 \quad (s^2 = 18.06)$$

$$\sigma = 4.03 \quad (\sigma^2 = 16.25) \quad \downarrow$$

B1B1

Under H_0 $\bar{X} \sim N \left(30, \frac{(4.25^2)}{10} \right)$

$$t = \frac{33.5 - 30}{4.25/\sqrt{10}} = 2.60$$

$$\frac{33.5 - 30}{4.03/\sqrt{9}} \quad (2.6 - 2.61)$$

M1A1

$$t_{\text{crit}} = 2.821$$

B1

do not reject H_0
 Insufficient evidence at the 1% level of significance that Jasmine's teacher is

underestimating the time that it takes to complete the homework assignments.

E1ft 7

(b) Times are Normally distributed

B1 1

[8]

Q8.

(a) $\bar{x} = 225.25$

B1

$s = 5.06$ ($s^2 = 25.6$)
($\sigma = 4.74$), ($\sigma^2 = 22.4$)

B1

$H_0 : \mu = 230$
 $H_1 : \mu \neq 230$

both

B1

$v = 8 - 1 = 7$

B1

$t_{crit} = \pm 2.365$

accept $t_{crit} = -2.365$

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B1

Test statistic:

$$t = \frac{225.25 - 230}{5.064 / \sqrt{8}} = -2.65$$

$$\frac{225.25 - 230}{4.74 / \sqrt{7}} = -2.65$$

M1

(-2.66 to -2.65)

A1

Reject H_0 at 5% level

A1ft

No evidence to support the producer's claim

E1ft

9

- (b) We have rejected H_0 when in fact H_0 may be true.
This indicates that a Type I error may have been made.

B2

2

[11]

Q9.

$$H_0: \mu = 1000$$

$$H_1: \mu \neq 1000$$

2-tailed test

$$\bar{x} = \frac{12036}{12} = 1003$$

$$S = 5.444$$

$$S^2 = 29.6$$

$$v = 12 - 1 = 11$$

$$t = \frac{\bar{x} - \mu}{S / \sqrt{n}} = \frac{1003 - 1000}{5.444 / \sqrt{12}} = 1.91$$

B1

B1

B1

B1

M1

A1ft

A1

$$t_{crit} = \pm 2.201$$

(on their v)

B1ft

Accept H_0

(on their t – values)

A1ft

Insufficient evidence to indicate a change

in the mean content of sherry in a bottle

E1ft

10

[10]

Q10.

(a) $\bar{x} = \frac{471}{5} = 94.2$

Or $s^2 = 36.7$

B1

$s = 6.058$

B1

$\nu = 4$
1-tailed test

B1

$t_{\text{crit}} = -2.132$

Or seen on diagram

B1

$H_0: \mu = 100$

$H_1: \mu < 100$

B1

$t = \frac{94.2 - 100}{\frac{6.058}{\sqrt{5}}} = -2.14$

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$\frac{\text{their } \bar{x} - 100}{\text{their } s / \sqrt{5}}$

M1A1

Reject H_0 at 5% level of significance
on their t and critical value

A1ft

Evidence at the 5% level of significance to support the members' belief that the batteries last less than 100 hours.

(t values only)

E1ft

9

(b) $\bar{x} = \frac{8080}{80} = 101$

$$s^2 = \frac{6399}{79} = 81 \quad s^2 = \frac{6399}{80} = 80.0$$

$$s = 9 \quad \text{or} \quad s = 8.944$$

For s (or s^2) and $\bar{x}$

B1

$$H_0: \mu = 100$$

$$H_1: \mu \neq 100$$

B1

$$\bar{X} \sim N\left(100, \frac{81}{80}\right) \text{ under } H_0$$

or 100, $\frac{9}{\sqrt{80}}$ used

B1

$$z = \frac{101 - 100}{\frac{9}{\sqrt{80}}} = 0.99$$

allow use of t method
(AWFW 0.99 to 1.00)

M1A1

2-tailed test

$$z_{\text{crit}} = \pm 1.96$$

on their z and critical value

B1

Accept H_0 at 5% level of significance

A1ft

sufficient evidence at the 5% level of significance to support the manufacturer's belief

E1ft

8

[17]

Q11.

$$H_0: \mu = 568$$

$$H_1: \mu < 568$$

$X \sim$ contents of cartons of milk

$$X \sim N(568, \sigma^2)$$

B1

1% one-tailed test

Under H_0 :

B1

$$v = 7$$

$$\bar{x} = \frac{4510}{8} = 563.75$$

$$\bar{X} \sim N\left(568, \frac{\sigma^2}{n}\right)$$

B1

$$\Rightarrow s^2 = \frac{254256.8}{7} - \frac{8}{7}(563.75)^2$$

$$s^2 = 7.929$$

$$(s = 2.816)$$

B2

$$t = \frac{563.75 - 568}{2.816 / \sqrt{8}}$$

M1

$$t = -4.27$$

$$(AWFW = 4.27 \text{ to } -4.26)$$

A1ft

$$t_{crit} = -2.998$$

B1ft

reject H_0

On their t

A1ft

Evidence at the 1% level of significance to suggest that the average contents of the cartons have been reduced.

E1ft

10

[10]

Q12.

(a) $H_0 \mu = 18$

One correct hypothesis – generous

B1

$H_1 \mu \neq 18$

Both correct – ungenerous

B1

$\bar{x} = 32.11$ $s = 18.71$
 $32.1 \text{ (} 32.05 \sim 32.15 \text{)}$

B1

$18.7 \text{ (} 18.65 \sim 18.75 \text{)}$

B1

Use of their s.d./√11

M1

$$t = \frac{\frac{32.11 - 18}{18.71}}{\frac{1}{\sqrt{11}}} = 2.50$$

Correct method for t ignore sign

m1

$2.50 \text{ (} 2.495 \sim 2.505 \text{)}$

A1

critical values t_{10} are ± 2.228

10df

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B1

± 2.228 their df, ignore sign, allow 2.23

B1ft

reject H_0 significant evidence mean not equal to (greater than) 18

Reject H_0 , must be compared with correct tail of t.

A1ft

10

(b) (i) $H_0 \mu = 18$ - no change

No change

B1

(ii) $H_1 \mu < 18$

$\mu < 18$ – generous

B1

- (iii) -1.812
 -1.812 cao

B1

- (iv) Accept H_0 mean equals 18
Correct conclusion based on correct answers to (i),(ii) and (iii)

B1

4

- (c) (i) $H_0 \mu = 18$ - no change
no change – allow $\mu <$

B1

- (ii) $H_1 \mu > 18$
 $\mu > 18$ generous

B1

- (iii) 1.812
 1.812 cao

B1

- (iv) Reject H_0 significant evidence mean greater than 18
Correct conclusion based on correct answers to (i),(ii) and (iii)

B1

4

Q13.

- (a) $\bar{x} = 54.44$ $s = 21.12$
 54.44 (54.4 – 54.5)

B1

21.12 (21.1 – 21.2) - may be implied

B1

- $H_0: \mu = 65$ $H_1: \mu < 65$
One correct hypothesis – generous

B1

Both hypotheses correct - ungenerous

B1

$$t = \frac{\frac{54.44 - 65}{21.12}}{\sqrt{9}} = -1.50$$

Use of their $\frac{s}{\sqrt{9}}$

M1

Correct method for t - ignore sign

m1

$$-1.50 (-1.49 - -1.51.)$$

A1

c.v. t_8 is -2.896

8.d.f.

B1

$$-2.896(-2.89 - -2.90) \text{ their d.f.}$$

B1 ft.

Accept H_0 , no significant evidence that mean is less than 65 seconds

Conclusion, their figures, must be compared with appropriate tail of t

$$\text{SC: } 65 - 2.896 \times \frac{21.12}{\sqrt{9}}$$

$$44.61(44.60 - 44.62) \text{ compare with } 54.44$$

$$54.44 + 2.896 \times \frac{21.12}{\sqrt{9}}$$

$$74.83(74.82 - 74.84) \text{ compare with } 65$$

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A1 ft.

10

(b) (i) No change

No change - requires correct null hypothesis in (a) or correctly stated null hypothesis here

B1

(ii) $\mu \neq 65$

$$\mu \neq 65$$

B1

(iii) ± 1.860

$$\pm 1.860 \text{ or } \pm 1.86$$

B1

(iv) no change

*No change - requires previous M1m1
A1 ft.*

A1 ft.

4

(c) (i) Conclude $\mu < 65$ when $\mu < 65$

Reject H_0 | H_0 true

E1

(ii) Conclude $\mu = 65$ when $\mu < 65$

Accept H_0 | H_0 untrue

E1

Both in context

E1

3

[17]

Q14.

(a) $\bar{x} = 14.3$ $s = 8.9621$
14.3 cao

B1

$H_0: \mu = 10$ allow $\mu \leq 10$
8.96 (8.96 – 8.97)

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B1

$H_1: \mu > 10$

*One correct hypothesis - generous
Both hypotheses correct - ungenerous*

B1 B1

$$t = \left(\frac{\frac{14.3 - 10}{8.9621}}{\sqrt{8}} \right)$$

*Use their $\frac{s.d.}{\sqrt{8}}$ – generous
Correct method for t– ignore sign*

M1 m1

$= 1.36$

$1.36 (1.35 - 1.365)$

A1 B1

c.v. t_7 is 1.895

$1.895(1.89 - 1.9)$ ignore sign

B1ft

Accept H_0 , no significant evidence to show mean exceeds 10 minutes

Conclusion, must be compared with correct tail of t. Allow 2 sided test.

A1ft

10

(b) $H_0: \mu = 10$

One correct hypothesis – ungenerous

B1

$H_1: \mu \neq 10$

Both hypotheses correct – ungenerous

B1

$$z = \frac{(11.21 - 10)}{\left(\frac{6.93}{\sqrt{190}}\right)} = 2.41$$

Method for z – ignore sign

$2.41 (2.4 - 2.41)$

M1 A1

c.v. ± 1.96

1.96 ignore sign, allow $t = 1.972 (1.97 - 1.975)$

B1

Reject H_0 , conclude mean $\neq$ (greater than) 10 minutes

Conclusion, must be compared with z or t, disallow if it is clear that wrong tail used.

Allow 1 tail test.

A1ft

6

(c) (i) Not at all – large sample – mean approximately normal

Not affected

Large sample / central limit theorem – mean not essential

B1 E1

- (ii) Conclusion unreliable – for example the company may have selected the most punctual trains
*Reason – bias is **not** enough – need example / explanation*

B1 E1

4

[20]

Q15.

- (a) $\bar{x} = 1.00$ $s = 4.266$
cao 4.26 - 4.27

B1

$H_0: \mu = 0$

one correct hypothesis

B1

$H_1: \mu \neq 0$

both hypotheses correct - must refer to population in words or symbols

B1

$t = (1 - 0) / (4.266 / \sqrt{8})$

*attempt to use $s/\sqrt{n}$
 correct method for t*

M1 M1

$= 0.663$

$0.66 - 0.664$

*cv from t-distribution
 7 df*

A1 B1 B1

$c.v. \pm 2.365$

*for ± 2.365 (2.36 – 2.37) ignore sign
 ft correct conclusion - must compare with correct tail of t-distribution*

B1 A1

10

Accept H_0 , i.e. accept that instrument is unbiased.

- (b) (i) $H_0: \mu = 0$

$H_1: \mu < 0$

alternative hypothesis

B1

$z = (-0.25 - 0)/(0.38/\sqrt{20}) = -2.94$
use of 0.38 /ft20
correct method for z -ignore sign
for -2.94 (-2.93 - -2.95)

M1 M1 A1

c.v. -2.3263
for -2.3263 (or -2.326 or -2.32 or -2.33)
ignore sign

B1

Reject H_0 , conclude second instrument underestimates on average.
correct conclusion, must be compared with correct
tail of normal distribution.

A1ft

6

- (ii) Evidence second instrument biased,
 no evidence first instrument biased
for comparison of bias/ size of errors

B1

Second instrument appears to be less variable.

B1

Second preferred because less variable,
 smaller errors on average and generally
 gives benefit of doubt to motorist.

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B1

3

[19]

Q16.

(a) $\bar{x} = 58.875$

$(58.87 - 58.9)$

B1

$s = 10.357$

$(10.3 - 10.4)$

both may be implied by correct t value

B1

$H_0 : \mu = 50$

One correct hypothesis

B1

$$H_1: \mu \neq 50$$

Both hypotheses correct

B1

$$t = \frac{(58.875 - 50)}{\left(\frac{10.357}{\sqrt{8}}\right)}$$

Use of t

Use of their s.d. / $\sqrt{8}$

Completely correct method

M1 M1 m1

$$= 2.42$$

$$(2.42 - 2.43)$$

A1

$$cv \quad t_7 \pm 2.365$$

7 degrees of freedom

B1

reject H_0

2.365 ignore sign

B1



mean not equal to 50 (> 50)

A1 ft needs both M1 marks; must be compared with correct (or both) tails of t distribution

A1ft

11

(b) random sample

B1

normal distribution

B1

- (c) (i) $H_0 : \mu = 48.8$
 $H_1 : \mu > 48.8$

Correct null hypothesis

B1

Correct alternative hypothesis

B1

$$z = \frac{(50.4 - 48.8)}{\frac{12.6}{\sqrt{89}}}$$

Use of $\frac{12.6}{\sqrt{89}}$

Completely correct method, ignore sign

M1 m2

$$= 1.20$$

$$1.20(1.19 - 1.21)$$

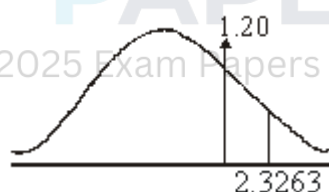
A1

cv 2.3263

2.3263 (2.32 – 2.33 or 2.368 – 2.371)

B1

Accept H_0



No significant evidence that mean has increased.

Correct conclusion - must be compared with correct tail of normal, allow 2-tail

A1ft

8

- (ii) Sample self-selecting - may be more ambitious / gullible students.
Non-random

E2

2

[23]

Examiner reports

Q2.

The hypotheses were generally well expressed in terms of μ although often $\neq$ was used in H_1 . Although the sign of the test statistic was usually correct, there was much confusion over the sign of the critical value. Sometimes a negative test statistic was compared with a positive critical value (often with a diagram showing that this was exactly what was being done). Frequently the critical value was stated as ± 1.6449 (sometimes with two-tailed diagrams) and from the two alternatives the negative was chosen because the test statistic was negative. The implication was that a test statistic of $+1.798$ would have been compared with $+1.6449$ and also resulted in the rejection of H_0 (as seemed to be the case where a modulus sign was used). The use of t_{19} rather than z was good in part (b). In part (c), most students appreciated the effect of the significance level in setting the probability of a Type I error.

Q3.

The great majority of candidates either did not state an assumption or simply stated incorrectly one of "It is Normal", "Sample is Normal", "Data are normally distributed" or "Shots are normally distributed". None of these responses gained any credit. Candidates should have stated "The parent population is / The lengths of shots are / The distances achieved are normally distributed". Almost all candidates found $\bar{x} = 184$ but their calculations of s or s^2 were less well done.

Some candidates did not seem to understand how to find the required value from the given information, with $s = \sqrt{1240} = 35.2$ seen too often. However, most candidates found the correct value of s^2 followed by a correct evaluation of the required t -statistic. Candidates should understand that, when the value of the population standard deviation is unknown and the sample size is small, then a t -test and **not** a z -test should be used, otherwise marks will be lost. Conclusions in context were sometimes either too positive or missing altogether.

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Q4.

The majority of candidates either did not state an assumption or simply stated incorrectly that "It is normal" or "Sample is normal" or "Data are normally distributed" or "Customers are normally distributed". None of these responses gained any credit. Candidates should have stated that "The parent population is normal" or "The number of customers is normally distributed". Almost all candidates found that $\bar{x} = 82$ and that $s = 5.58$ (or $s^2 = 31.1$).

The correct evaluation of the required t -statistic (1.86) was often seen. However, some candidates appeared not to understand that, when the value of the population standard deviation is unknown and the sample size is small, then a t -test and **not** a z -test must be used. As a result, too many lost a significant number of marks. Conclusions in context were sometimes either too positive or missing altogether.

Q5.

Many excellent solutions were seen to part(a)(i), with $\bar{x} = 43.5$, $s = 2$ and $t = 2.365$ often

used in the calculation of the 95% confidence interval as $43.5 \pm 2.365 \times \frac{2}{\sqrt{8}}$. Some candidates stated $s = 1.87$ or $s^2 = 3.5$, which only gained credit if used correctly in $43.5 \pm 2.365 \times \frac{1.87}{\sqrt{7}}$.

The vast majority of candidates either did not state an assumption or simply stated “It is normal” or “Sample is normal” or “Boxes are normally distributed”. None of these gained any credit as it was the parent population or the weights of boxes of black peppercorns that were assumed to be normally distributed. In part (a)(ii), the comment was either too positive with candidates stating “Bishen’s belief is true” or no reason was given as to why Bishen’s belief may or may not have been justified.

In part (b)(i), the hypotheses were usually stated correctly, followed by a correct evaluation of the test statistic as -2.12 . However, far too many candidates stated that the critical value was $t = +1.895$ or $t = \pm 1.895$ and consequently went on to consider the wrong tail of the distribution, with the inevitable loss of credit. Part (b)(ii) was usually done very well, with only the occasional incorrect mention of a Type II error.

Q6.

Most candidates apparently assumed that they had a **small** sample from a normal distribution with **unknown** variance and so used the t -distribution when conducting the required test. However, many such candidates either did not give an assumption or apparently did not know what assumption they had actually made. The assumption that “The sample is normally distributed” was the usual false statement and “It is normally distributed” or simply “Normally distributed” were also insufficient to gain the mark available. It was the **population** from which the sample was taken that had to be assumed to have a normal distribution. As in Question 3, hypotheses were sometimes either incorrectly stated or not stated at all. The correct values of $\bar{x} = 4.1$ and $s = 0.392$ were usually found and correct conclusions, in context, were usually seen.

Q7.

This question proved to be another good source of marks with the majority of candidates gaining at least 7 marks. Part (a) was done by a variety of correct methods, with good conclusions in context often seen. Most candidates realised that a t -test was required, with only a few weaker candidates wrongly using $z = 2.3263$, instead of $t = 2.821$, as their critical value. In part (b), it was the population of times taken to complete the French assignment that must be assumed to be normally distributed and not, as often quoted, the sample itself.

Q8.

The values for $\bar{x}$ and s were usually found correctly. The comments regarding hypotheses, sketches and conclusions, as detailed for question 3, are equally valid here. Many candidates used z -values (usually ± 1.96) when t -values were required because the test involved a small sample from a normal distribution with unknown variance. This was a 2-tailed test and, as such, conclusions had to consider whether or not there was a change (not a decrease) from 230 grams in the mean weight of jam.

Answers to part (b) were either correct or incorrect. Some candidates simply stated “Type I error” without any evidence in context to back up their claim, and consequently received

no credit. Others incorrectly stated that there was a Type II error because they had rejected H_0 when, in fact, H_0 was true.

Q9.

There were many excellent solutions seen to this question. Most candidates stated the correct hypotheses and then attempted to use a 2-tailed t -test. The main errors occurred in either looking up the wrong t -value from the tables or not giving a conclusion in context.

Q10.

In general, candidates scored well on this question, with most attempting to state the hypotheses correctly and then giving conclusions in context based on their findings. However, such conclusions were often too positive. In part (a), most candidates realised that a small sample from a normal distribution with unknown variance required the use of a t -distribution. Unfortunately there were too many candidates who failed to distinguish which tail they were using, simply stating $t_{crit} = 2.132$ from tables without then applying $t_{crit} = 2.132$ to the left-hand tail. As stated in previous reports, a diagram is often very useful in such questions to demonstrate the critical values and acceptance or rejection regions. Calculating the standard deviation from the information provided in part (b) of the question seemed to be problematic for many candidates; this despite the relevant formula being given on page 12 of the formulae booklet!

Q11.

The null and alternative hypotheses were usually correctly stated and values for $\bar{x}$ and s (or s^2) found correctly to be 563.75 and 2.816 (or 7.929) respectively. The majority of candidates realised that a small sample from a population with unknown variance suggested a t -distribution. Only a few candidates, used z -values here. However, there are still some candidates who seem to have very little idea as to which of the tables to use from the supplied booklet. Although candidates, on the whole, attempted to give a conclusion in context, they were often far too positive in nature and/or did not make any reference to the 'average contents' of cartons.

Q13.

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Part (a) was well answered. In particular, it was pleasing to see that most candidates realised that both the test statistic and the critical value were negative. As in question 1, a few candidates lost marks by failing to make clear, in words or symbols, that the hypotheses applied to the population. Relatively few gained full marks for part (b), often suggesting that the conclusion would change in part (iv).

Q14.

Candidates were well prepared for this question and generally coped well with the perhaps unfamiliar "not more than" claim in line 1. Calculations were usually accurate and most candidates gave sensible answers to part (c).

Q15.

As in question 5 most candidates seemed well prepared for part (a), although there were more errors in calculating the mean and standard deviation. This was presumably because of a difficulty in coping with negative values from the t -distribution.

It was heartening to see in part (b) that most candidates correctly obtained a negative

value for z and correctly compared it with a negative critical value.

Q16.

Candidates were well prepared for this question. The calculations provided few problems and the majority of candidates made clear in words or, more commonly, in symbols that their hypotheses referred to the population. In part (c), the comments made in the previous report (January 2002) concerning large samples had clearly been heeded.

There was a substantial number of correct answers to part (c)(ii), although many answers could be summarised as 'a Type 1 error might occur'. This latter had been specifically ruled out as a valid answer to the question.



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